

Arc Reactor Mining Lab - 2026-05-29

Status: PDF-ready package candidate.

This report is intentionally a four-part package specification. It is not a profit promise and not a miner installer.

PAGE 1 - EXECUTIVE SUMMARY AND REPO TOUCHPOINTS

Arc Reactor Mining Lab is a local-first, legal, operator-controlled package inside alex-place/lantern-os. Its strongest version inventories the fleet, routes hardware only into viable lanes, validates wallets and claims in read-only mode, and emits receipts for tiny compliant sell-to-USD tests.

One shortcut rule: Lantern OS remains the front door. Mining Lab is an internal card backed by real files, not a new dashboard or launcher.

Touchpoint	Role
skills/solo-mining/SKILL.md	Skill contract and hard safety blocks
docs/ARC-REACTOR-MINING-LAB.md	Operator guide
templates/*.csv and templates/mining-receipt.json	Intake, wallet, feasibility, and receipt templates
scripts/Get-HardwareInventory.ps1	Windows inventory collector
scripts/Test-MiningProfitability.ps1	Formula implementation
scripts/Install-LanternShortcut.ps1	Single Lantern shortcut
surfaces/tony-garage/index.html	Backed Mining Lab card

Diagram:

```
Inventory
|
Wallets - Arc Reactor - Claims
|
Mining Lanes
|
Receipts
|
Tiny USD Seed
```

PAGE 2 - COIN FEASIBILITY AND FORMULA

The correct feasibility model is not which coin the fleet can dominate. It is which lane matches owned hardware, power price, heat/noise tolerance, off-ramp availability, and operator risk.

Coin	Hardware lane	Verdict
BTC	SHA-256 ASIC	Reject for CPU/GPU fleet
ETH	None	Wallet, claim, and microbuy only
XMR	CPU	Best learning lane
RVN	GPU	Experiment only
ETC	GPU or ETChash ASIC	Sunk-cost hardware only
LTC	Script ASIC	ASIC only
DOGE	Script ASIC	Usually evaluated with LTC economics
KAS	kHeavyHash ASIC	Owned ASIC only

Universal model:

```
coins_per_day = (effective_hashrate / network_hashrate) * block_reward * blocks_per_day * (1 - pool_fee)
gross_usd_day = coins_per_day * spot_price_usd
power_usd_day = (watts / 1000) * hours_per_day * power_usd_per_kwh
net_usd_day = gross_usd_day - power_usd_day
```

Power and profitability fields must be refreshed before every decision. Static ROI claims are rejected.

PAGE 3 - WALLET MATRIX AND READ-ONLY CLAIMS

The wallet layer is the durable value of the lab. It can hold earned dust, testnet balances, approved microbuys, watch-only balances, and later off-ramp receipts without turning Lantern OS into a custody tool.

Lane	What it is for	Boundary
Watch-only checks	BTC/ETH balance evidence	No private keys
Pool/P2Pool mining	Owned hardware rewards	No unauthorized devices
Testnet/faucets	Practice and screenshots	Not revenue
EVM claim guard	Selector review	No blind signing
Microbuys	Real tiny rows	Documented lawful venue
Receipt ledger	Proof of process	Append-only corrections

Read-only scripts:

```
skills/solo-mining/examples/read_only_eth_balance.py
skills/solo-mining/examples/read_only_btc_balance.py
skills/solo-mining/examples/claim_guard.py
```

Claim guard blocks 0x095ea7b3 for ERC-20 approvals and 0xa22cb465 for ERC-721 setApprovalForAll. Those operations can authorize other parties to move assets, so they require explicit human review.

PAGE 4 - COMMERCIAL PACKAGE, TIMELINE, ACCEPTANCE CRITERIA, SOURCES

The sellable offer is not "buy a miner and get rich." The offer is: "I will tell you what your hardware can actually do, set up safe watch-only wallet checks, and stop you from wasting money on impossible mining lanes."

Offer	Deliverables	Indicative price
Arc Reactor Mining Intake	hardware intake, wallet matrix, feasibility note	\$49
Arc Reactor Mining Lab Lite	intake, wallet matrix, read-only scripts, profitability sheet	\$149
Arc Reactor Mining Lab Full	Lite plus shortcut, report, off-ramp checklist	\$249
Monthly receipt audit	refreshed model, watch-only audit, receipt archive	\$29-\$49/mo

Acceptance criteria:

Area	Pass condition
Skill boundary	Blocks ETH mining, brute force, unauthorized transfers, hidden signing, fake ROI
Inventory	Writes non-empty CPU/GPU/RAM output and allows manual ASIC rows
Profitability	Computes gross, power cost, and net for a fixed fixture
ETH monitor	Uses eth_getBalance only
BTC monitor	Uses watch-only address APIs only
Claim guard	Rejects approve and setApprovalForAll selectors
Surface honesty	Mining Lab card links only to real files
Shortcut discipline	One Lantern shortcut; optional diagnostics only
Repo scope	No first-release gm-agent-orchestrator code changes
Reporting	Pilot emits receipt JSON and markdown report

Sources: Ethereum proof-of-stake and JSON-RPC docs, Monero RandomX docs, Ravencoin official KAWPOW page, Ethereum Classic mining guide, Kaspero homepage, Blockstream Esplora API, EIP-20, EIP-721, EIA electricity data, and U.S. Treasury digital asset broker reporting releases.